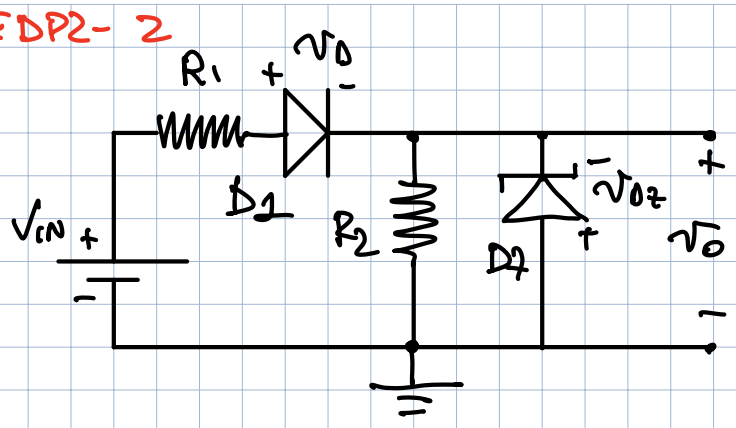
EDP2- 2



$R_1 = 1K\Omega$ $R_2 = 3K\Omega$

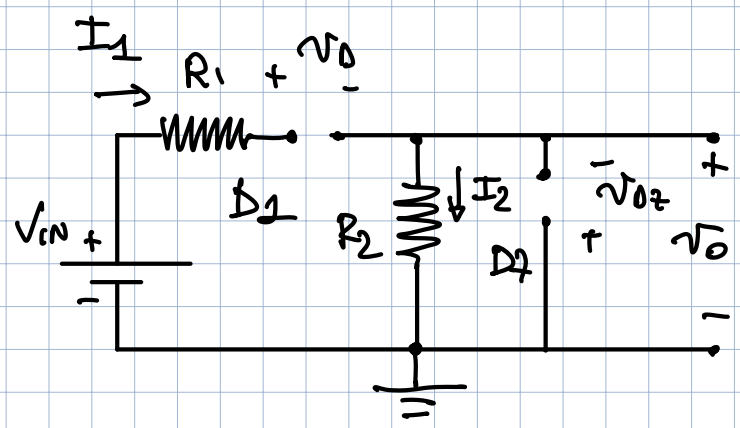
$V_{ON} = 1V$ $V_Z = 7{,}5V$

TROVARE v_O QUANDO

$V_{IN} = -5V$
$V_{IN} = 5V$
$V_{IN} = 40V$

FACOLTATIVO! TROVARE $v_O = f(V_{IN})$

HP D1 e D2 = OFF



CIRCUITO APERTO

$I_1 = 0$, $I_2 = 0$
$V_O = 0$

$v_{D1} = V_{IN} - V_O = V_{IN}$

$v_{DZ} = 0$

1) $V_{IN} = -5V$

$\Rightarrow v_{D1} = -5V \Rightarrow D_1 = OFF$

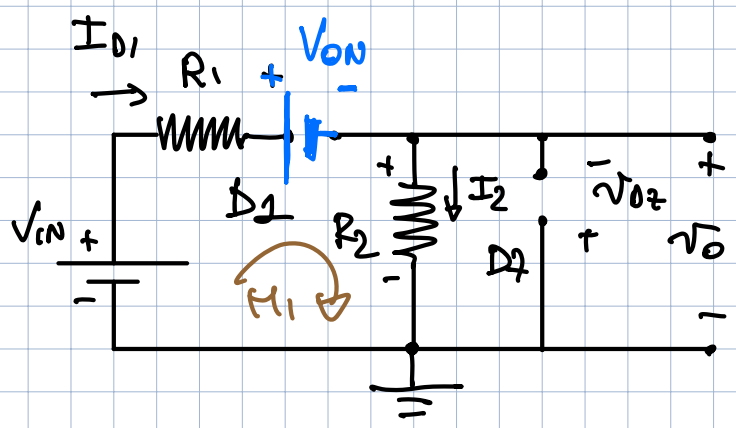
$v_{DZ} = 0 < V_{ON} \Rightarrow D_Z = OFF$ | VERIFICATO

$\Rightarrow$ $V_O = 0V$

2) $V_{IN} = 5V$ (RISPETTO AL CASO D1 = D2 = OFF SI HA:)

$v_{D1} = +5V \Rightarrow D_1 = ON$

$v_{DZ} = 0V \Rightarrow D_Z = OFF$



M_1: $I_{D1} = I_2$

$$-V_{IN} + R_1 I_{D1} + V_{ON} + R_2 I_{D1} = 0$$

$$I_{D1} = \frac{V_{IN} - V_{ON}}{R_1 + R_2} = \frac{4V}{4K\Omega} = 1mA > 0$$

OK $D_1 = ON$

$$v_{DZ} = -R_2 I_2 = -R_2 I_{D1} = -3K\Omega \cdot 1mA = -3V$$

$$-7{,}5V \leq v_{DZ} \leq 1V \Rightarrow \text{OK } D_Z = OFF$$

$$\Rightarrow v_O = I_2 R_2 = 3V$$

2) $V_{IN} = 40V$ (RISPETTO AL CASO $D_1 = D_Z = OFF$ SI HA:)

$v_{D1} = +40V \Rightarrow D_1 = ON$

$v_{DZ} = 0V \Rightarrow D_Z = OFF$

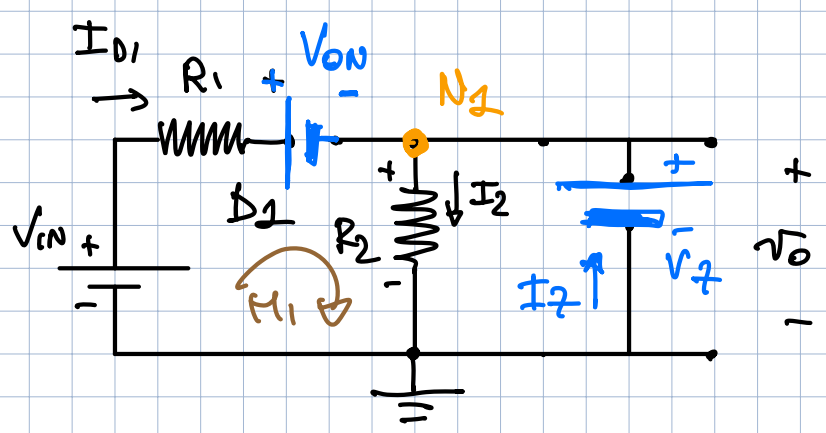
IL CIRCUITO È LO STESSO DEL CASO PRECEDENTE E TROVO:

$$I_{D1} = \frac{V_{IN} - V_{ON}}{R_1 + R_2} = \frac{39V}{4K\Omega} = 9{,}75mA$$

$$v_{DZ} = -R_2 I_{D1} = -29{,}25V$$

$v_{DZ} < -V_Z \Rightarrow$ D_Z NON È OFF MA È IN BREAKDOWN!

$\Rightarrow$ QUINDI CONSIDERO $D_1 = ON$ e $D_Z = BREAK$



$V_o = V_Z = 7{,}5V$

DEVO VERIFICARE CHE $I_{D1} > 0$

$I_Z < 0$

$$I_{D1} = \frac{V_{IN} - (V_{ON} + V_Z)}{R_1} = \frac{40V - 8{,}5V}{1K\Omega} = 31{,}5\ mA > 0$$

OK D1 ON

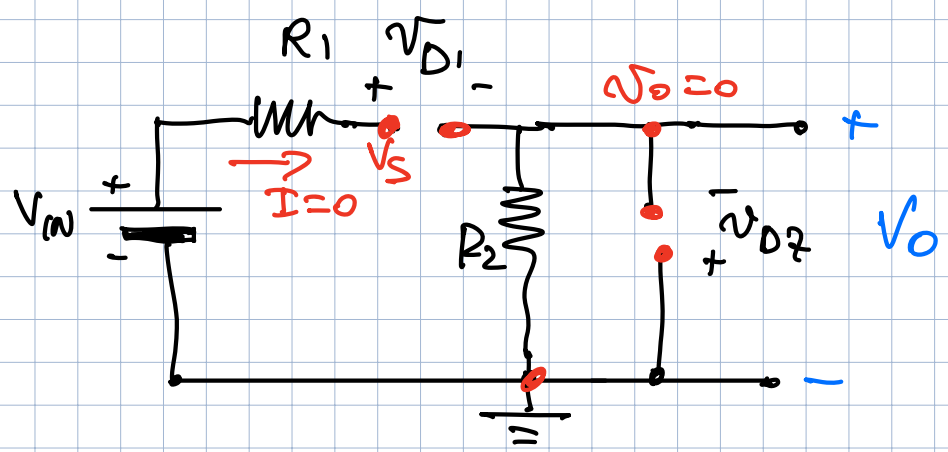
$$I_2 = \frac{V_Z}{R_2} = \frac{7{,}5V}{3K\Omega} = 2{,}5\ mA$$

N_1: $I_{D1} + I_Z - I_2 = 0$ $\quad I_Z = I_2 - I_{D1} = -28\ mA < 0$

OK DZ = BREAK

FACOLTATIVO! TRANSCARATTERISTICA

HP DIODI OFF



$v_o = 0$

$v_{D1} = V_{IN}$

$v_{DZ} = 0$

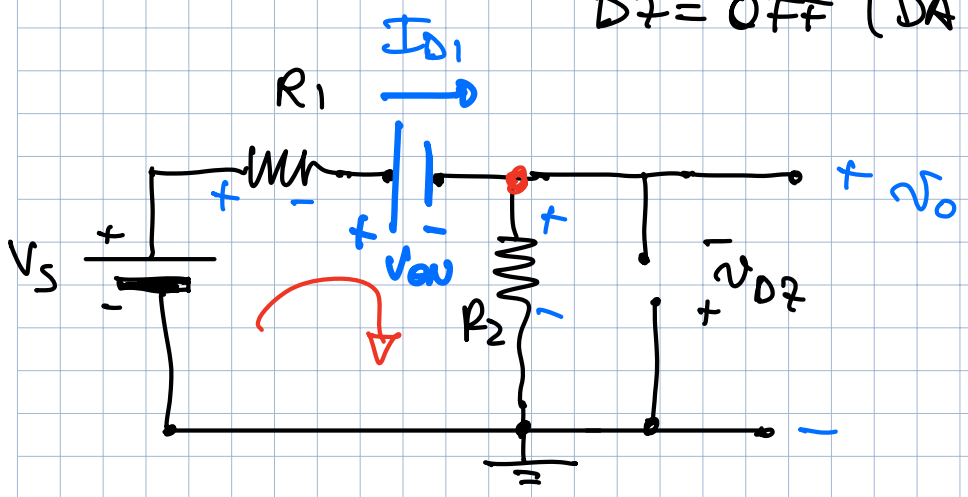
CON $\underbrace{v_{D1} < V_{ON}}_{V_{TH1}}$ ALLORA D = OFF e DZ = OFF

e $v_o = 0$

$\Rightarrow v_{D1} < V_{ON} \Rightarrow$ D1 e DZ OFF e $V_o = 0$

CON $V_{IN} > V_{ON}$ $\Rightarrow$ D1 = ON

D2 = OFF (DA VERIFICARE)



$$I_{D1} = \frac{V_S - V_{ON}}{R_1 + R_2} > 0$$

PER AVERE

$I_{D1} > 0 \Rightarrow V_S > V_{ON}$

OK D = "ON"

$$v_o(v_{IN}) = V_S \frac{R_2}{R_1+R_2} - V_{ON}\frac{R_2}{R_1+R_2}$$

$$v_o = I_{D1} R_2$$

$$= V_S \cdot \frac{3}{4} - 1V \cdot \frac{3}{4} = \boxed{0{,}75 V_S - 0{,}75 V}$$

RACCORDO $V_o(V_{ON}) = V_o(1V) = 0V$ OK

$$v_{D2} = -v_o = -0{,}75 V_S + 0{,}75 V$$

AFFINCHÉ D2 SIA "OFF" e NON IN BREAKDOWN

$$-V_Z \le v_{D2} \le V_{ON}$$

con $V_{IN} > 1V \Rightarrow v_{D2} < V_{ON} \Rightarrow$ OK D2 OFF

MA DEVE ESSERE $v_{D2} > -V_Z$

$$-0{,}75 V_S + 0{,}75 > -V_Z$$

$$\Rightarrow V_{IN} < \frac{V_Z + 0{,}75V}{0{,}75V} = \frac{7{,}5V + 0{,}75}{0{,}75} = 11V$$

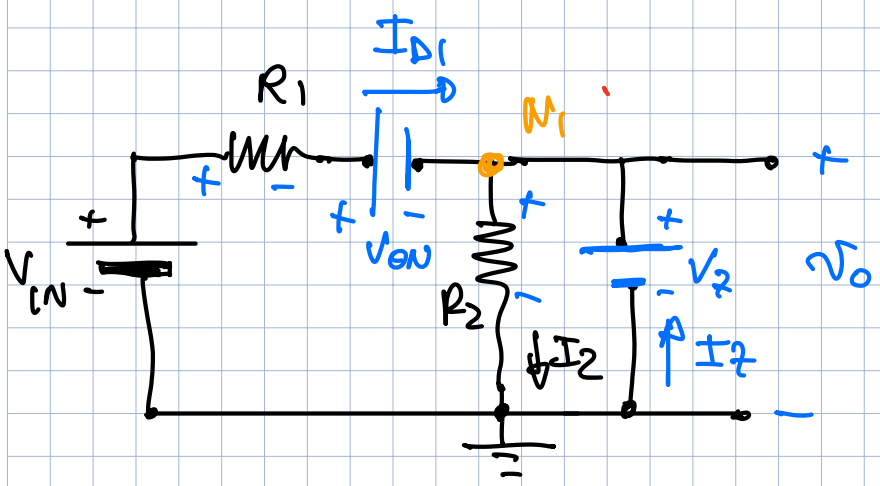
$V_o(11V) = 7{,}5V$

QUINDI

$$\boxed{\begin{array}{l} V_{ON} \le V_{IN} \le 11V \\ V_o = 0{,}75 V_{IN} - 0{,}75 \end{array}}$$

CON $V_{IN} > 11V \Rightarrow D_Z =$ BREAKDOWN

$D_1 =$ ON DA VERIFIC.



$v_O = V_Z = 7{,}5V$

(OK RACCORDO)

$$I_{D1} = \frac{V_{IN} - V_{ON} - V_Z}{R_1} > 0$$

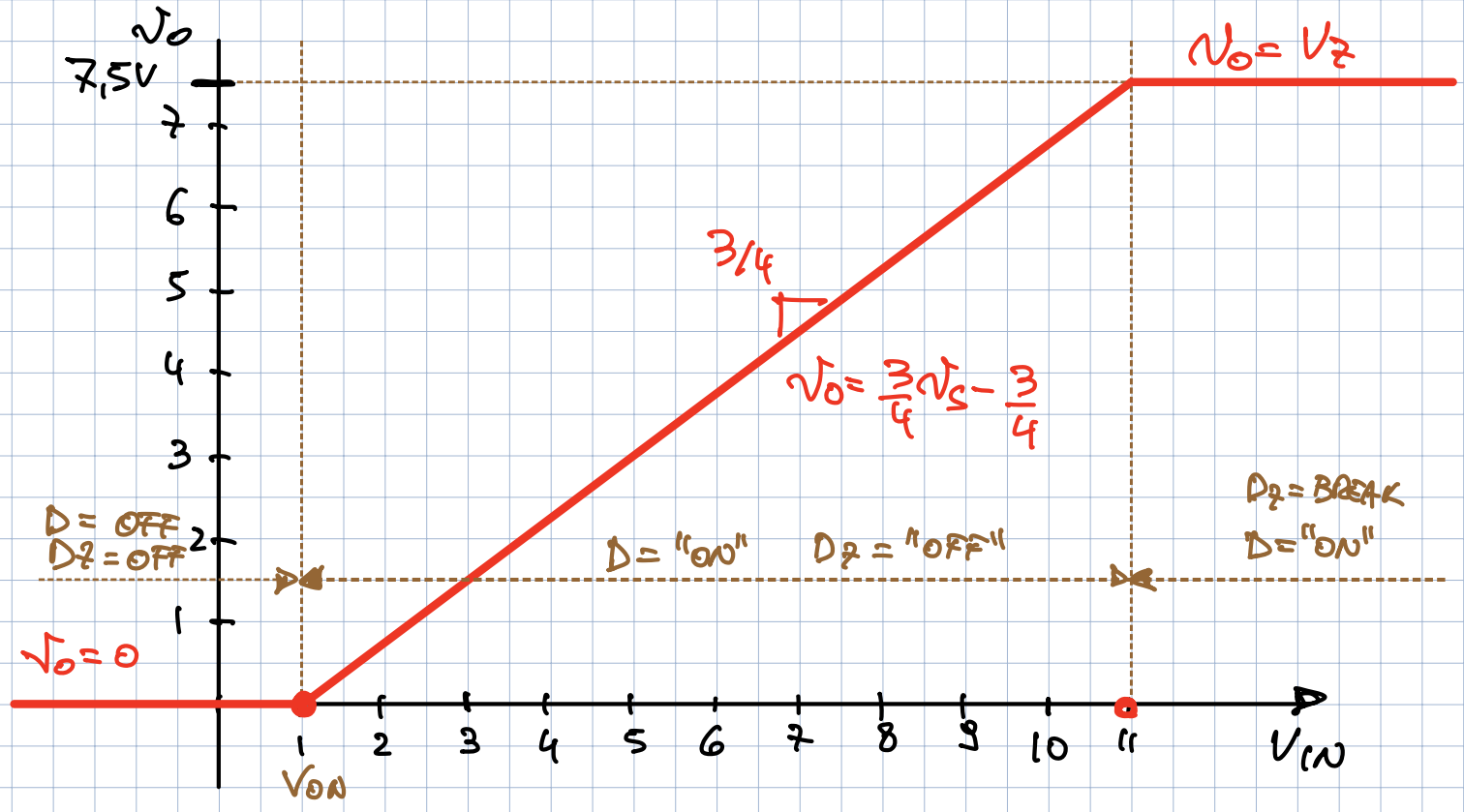
$V_{IN} > V_Z + V_{ON} = 8{,}5V$

OK xché SIAMO PER $V_{IN} > 11V$!

$I_{DZ} < 0$!

Su N_1: $I_{D1} + I_Z - I_2 = 0$

$$I_Z = I_2 - I_{D1} < 0 \Rightarrow \frac{V_Z}{R_2} - \left(\frac{V_{IN}}{R_1} - \frac{V_{ON} + V_Z}{R_1}\right) < 0$$

$$\frac{V_{IN}}{R_1} > \frac{V_Z}{R_2} + \frac{V_{ON} + V_Z}{R_1} \qquad V_{IN} > \frac{V_Z}{3} + 8{,}5V = 11V$$

OK